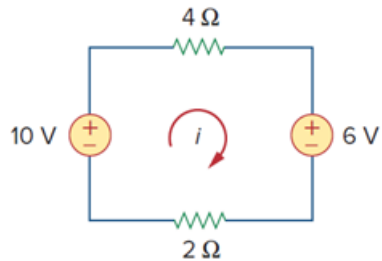


Problem

The current i in the circuit of Fig. 3.48 is:

- (a) -2.667 A
- (b) -0.667 A
- (c) 0.667 A
- (d) 2.667 A

Figure 3.48



Step-by-step solution

Step 1 of 3

Refer to Figure 3.48 in the textbook.

Determine the current i in the circuit.

Step 2 of 3

Apply KVL in Figure 3.48.

$$-10 + 4i + 6 + 2i = 0$$

$$-4 + 6i = 0$$

$$6i = 4$$

$$i = \frac{4}{6}$$

$$i = 0.667 \text{ A}$$

Hence, options (a), (b), and (d) are incorrect.

Step 3 of 3

The current in the circuit i is 0.667 A.

Hence, the correct option is **(c)**.